WORKSHEET 6.4

Accumulation Functions and the Fundamental Theorem, Part 1

Unit 6: Integration and Accumulation of Change

Key ideas

- **Fundamental Theorem, Part 1.** If f is continuous, then $\dfrac{d}{dx}\displaystyle\int_a^x f(t)\,dt = f(x)$.
- With a function as the upper limit, use the chain rule: $\dfrac{d}{dx}\displaystyle\int_a^{g(x)} f(t)\,dt = f\big(g(x)\big)\,g'(x)$.
- If the variable is the *lower* limit, the sign changes.
- An accumulation function is zero where its two limits are equal: $\displaystyle\int_a^a f(t)\,dt = 0$.

Worked examples

Example 1 · The basic case

Find $\dfrac{d}{dx}\displaystyle\int_2^x \sqrt{1+t^4}\,dt$.

Solution. Replace t by x in the integrand.

$\boldsymbol{\sqrt{1+x^4}}$**.**

Example 2 · With the chain rule

Find $\dfrac{d}{dx}\displaystyle\int_0^{x^3} \sin t\,dt$.

Solution. Substitute the upper limit and multiply by its derivative.

$\boldsymbol{3x^2\sin(x^3)}$**.**

Example 3 · Values of an accumulation function

Let $g(x) = \displaystyle\int_1^x \left(3t^2 - 2\right)\,dt$. Find $g(1)$, $g'(2)$ and $g''(x)$.

Solution. Equal limits give $g(1) = 0$. $g'(x) = 3x^2 - 2$.

$\boldsymbol{g(1) = 0}$**,** $\boldsymbol{g'(2) = 10}$**,** $\boldsymbol{g''(x) = 6x}$**.**

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Practice

CORE

1. Find $\dfrac{d}{dx}\displaystyle\int_0^x \cos\left(t^2\right)\,dt$.

2. Find $\dfrac{d}{dx}\displaystyle\int_5^x \frac{e^t}{t}\,dt$.

3. Let $F(x) = \displaystyle\int_3^x \left(t^2+1\right)\,dt$. Find $F(3)$.

4. For the function F in Question 3, find $F'(3)$.

EXAM LEVEL

5. Find $\dfrac{d}{dx}\displaystyle\int_x^4 \sqrt{t+1}\,dt$.

6. Find $\dfrac{d}{dx}\displaystyle\int_0^{2x} \ln(1+t)\,dt$.

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7. Find $\dfrac{d}{dx}\displaystyle\int_x^{x^2} t^3\,dt$.

8. Let $h(x) = \displaystyle\int_0^{\sin x} e^{t^2}\,dt$. Find $h'(0)$.

Question 9 · Free-response practice

Let $F(x) = \displaystyle\int_0^x \left(t^2 - 4t\right)\,dt$ for $x \geq 0$.

(a) Find $F'(x)$.

(b) On what interval is F decreasing? Justify your answer.

(c) Find the x-coordinate of the point of inflection of the graph of F, and the value of $F(4)$.

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Worked solutions

1. $\cos(x^2)$.
2. $\dfrac{e^x}{x}$.
3. 0.
4. $F'(x) = x^2 + 1$, so $F'(3) = 10$.
5. $-\sqrt{x+1}$.
6. $2\ln(1+2x)$.
7. $(x^2)^3(2x) - x^3 = 2x^7 - x^3$.
8. $h'(x) = e^{\sin^2 x}\cos x$, so $h'(0) = 1$.
9. (a) $F'(x) = x^2 - 4x$.
   (b) $F'(x) = x(x-4) < 0$ for $0 < x < 4$.
   (c) $F''(x) = 2x - 4$ changes sign at $x = 2$. $F(4) = \dfrac{64}{3} - 32 = -\dfrac{32}{3}$.

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